

Microsoft Fabric – Learning project scope

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OVERVIEW

1. Project background and description

This project builds a unified analytics platform on Microsoft Fabric that integrates mobility data (NYC Taxi), environmental data (OpenAQ air quality), and macro-economic data (GDP & FX) into a single data lakehouse and warehouse.

By combining these datasets, we can explore how urban mobility, environmental quality, and economic trends intersect. The solution uses Microsoft Fabric's Lakehouse, Data Factory, Dataflows, Notebooks, and Warehouse with a medallion architecture (Bronze → Silver → Gold) and delivers dashboards for decision makers.

Prerequisites:

Create a Fabric workspace and add:

- Lakehouse
- Data Factory → Dataflow Gen2 (for APIs) and Pipelines (for file copy/orchestration)
- Warehouse
- Notebook (PySpark)
- Power BI dataset/report (optional)

2. Project goals

1. Data Unification

Ingest and standardize multiple heterogeneous data sources (Parquet, JSON APIs, CSV).

2. Cross-Domain Analytics

Enable correlation between taxi usage, air pollution levels, and economic trends.

3. Actionable Insights

Provide insights for urban planning, transport policy, and sustainability.

4. Demonstrate Fabric Capabilities

Showcase ETL pipelines, Delta Lake, Warehouse star schema using open-source data.

DATA SOURCES & POTENTIAL INSIGHTS

3. Datasets

NYC Taxi Trips (Mobility)

- Data: trip pickups/drop-offs, fares, distance, passenger counts.
- Insights:
 - o Peak travel times and zones (rush hour, airports).
 - o Revenue and trip duration trends.
 - o Seasonal patterns in urban mobility.
- Source: official TLC (Taxi and Limousine Commission) page with monthly Parquet links (e.g., CloudFront URLs like `.../trip-data/yellow_tripdata_2025-06.parquet`). Files are published monthly with ~2-month lag.
- Link: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>.

OpenAQ (Air Quality)

- Data: hourly/daily measurements of pollutants (PM2.5, NO2, O3, etc.).
- Insights:
 - o Identify pollution hotspots by location/time.
 - o Correlate high-traffic zones with air quality degradation.
 - o Detect seasonal air quality variations.
- Source: API returns measurements; pagination via page and limit (max 1000 / page). Land JSON→tabular via Dataflow Gen2 to Lakehouse Tables (Delta).
- Link: <https://docs.openaq.org/about/about>

World Bank GDP & ECB FX (Economy)

- Data: yearly GDP per country (USD), daily USD/EUR FX rates.
- Insights:
 - o Contextualize transport revenues vs macro indicators.
 - o Convert taxi revenues into EUR to study currency impact.
 - o Show long-term growth vs. environmental/mobility metrics.
- Sources and Links: Macroeconomics (World Bank) + FX (ECB)
 - o World Bank GDP (e.g., USA GDP, JSON):
<https://api.worldbank.org/v2/country/USA/indicator/NY.GDP.MKTP.CD?format=json>
 - o ECB daily FX (USD/EUR, CSV):
<https://data-api.ecb.europa.eu/service/data/EXR/D.USD.EUR.SP00.A?format=csvdata>

4. Key analytical questions

- How does traffic intensity (trips per day) relate to air quality (PM2.5/NO2 levels) in NYC?
- Which zones or times of day show the strongest link between taxi demand and pollution peaks?
- What is the average revenue per trip in USD vs EUR, and how does exchange rate fluctuation affect it?
- Over multiple years, do we see mobility/economic growth at the expense of environmental quality?

PROJECT TASKS

Phase 1: Data Ingestion (Bronze)

- Set up Data Pipelines for NYC Taxi monthly Parquet files.
- Create Dataflows Gen2 for OpenAQ, World Bank, and ECB.
- Land all raw data in OneLake (Bronze).

Phase 2: Data Transformation (Silver)

- Standardize schemas (PySpark in Fabric Notebooks).
- Deduplicate, normalize, and enrich with date/time dimensions.
- Write cleaned Delta tables into Silver Lakehouse.

Phase 3: Data Modeling (Gold)

- Build a star schema in Fabric Warehouse:
 - FactTaxiDaily (daily trips, fares).
 - FactAirQualityDaily (daily pollutants per city/zone).
 - DimDate, DimZone, DimFX, DimGDP.
- Aggregate measures (daily totals, averages).

Phase 4: Analytics & Visualization

Power BI (optional) or Notebook visualizations:

- Mobility Dashboard: trips/day, average fare, busiest pickup zones.
- Air Quality Dashboard: PM2.5/NO2 trends by day and location.
- Mobility vs Air Quality Correlation: overlay taxi trips with pollution spikes.
- Economic Impact Dashboard: revenue in USD vs EUR, GDP context.

Phase 5: Governance & Monitoring

- Enable Purview lineage in Fabric for data governance (optional).
- Implement Row-Level Security (RLS) in Power BI for role-specific views (optional).
- Automate refresh schedules (daily/hourly).

5. Deliverables

- Fabric Lakehouse (Bronze, Silver, Gold) with three datasets.
- ETL Pipelines & Dataflows for automated ingestion.

- Fabric Warehouse with star schema for reporting.
- Visualizations with mobility, environment, and economy insights (Power BI optional).
- Documentation: data dictionary, lineage diagrams, governance policies.

6. Expected outcomes

- Unified view of how urban mobility affects air quality.
- Ability to convert taxi economics into global terms (GDP, FX) for policy-level insights.
- Demonstration of Microsoft Fabric's full stack for a data engineering and analytics project.
- A replicable framework to extend with other open datasets (e.g., weather, traffic sensors, energy).

7. Additional

Please, pay attention to the last point (extendability). Additionally add the following external "integrations":

1. Add weather data. Then implement an external job that periodically gets and copies enriched weather data into a separate time-series database (you may choose any). Build a dashboard to analyze the data (e.g. Grafana).
2. Use Great Expectations to define data quality checks (like distributions, nulls, etc.) on data and make a user-friendly trigger – sending a message to the bot in telegram or discord – to produce a report on quality of data and failures.